MagViVit

Transformer-Based Spatiotemporal Magnification for Vital Signs Monitoring

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Introduction

- Preterm infants in the NICU need continuous vital sign monitoring (RR, Temperature).
- Adhesive sensors cause skin injuries (MARSI), stress, and detach frequently.
- Goal: Non-contact vital sign prediction from NIR video using deep learning.
- Eliminates skin contact, camera replaces physical sensors.
- Clinical site: Soroka Medical Center NICU, Beer-Sheva.

Technical Objectives

- RR target: MSE ≤ 2.0 breaths/min (normal range: 30–60 bpm).
- Temperature target: MSE ≤ 0.2 °C (normal range: 36.5–37.5 °C).
- Dual-branch: CNN spatial features + Transformer temporal modeling.
- Ground truth from classical CV monitor extraction.

Algorithms & Design Considerations

Pre-processing

NIR → Torso & Face Detection → SORT tracking

RR Branch

Motion Magnification → Grayscale → ResNet-50 → Transformer Encoder.

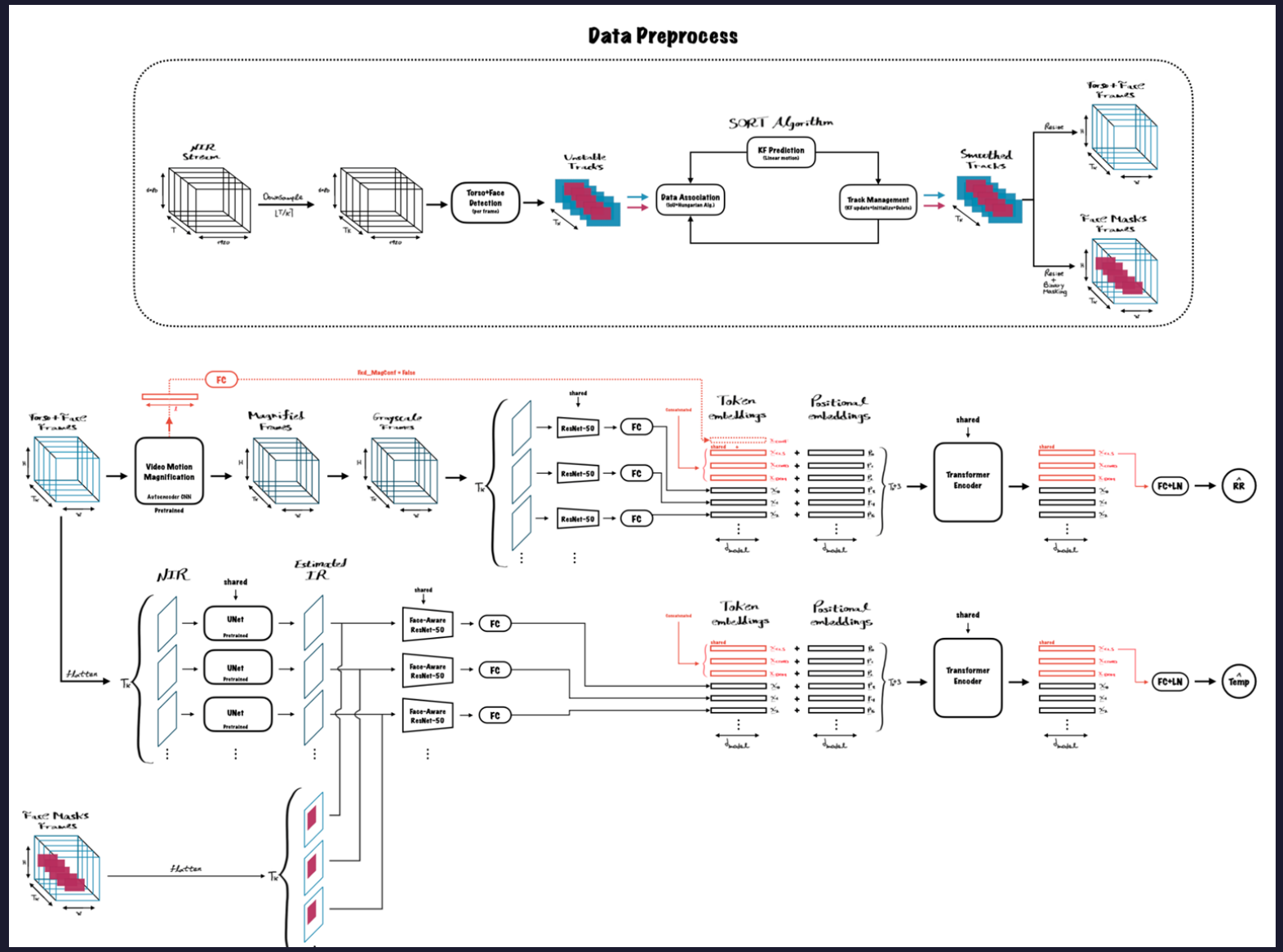
- Autoencoder CNN amplifies sub-pixel chest movements from NIR video.
- ResNet-50 extracts spatial features from magnified frames.
- Transformer models temporal breathing patterns (0.5–1.5 Hz).

Temperature Branch

NIR-to-IR UNet → Face-aware ResNet-50 → Transformer Encoder.

- UNet converts NIR to estimated IR thermal maps.
- Face-aware ResNet-50 focuses on facial thermal ROI.
- Transformer captures slow temperature dynamics over longer windows.

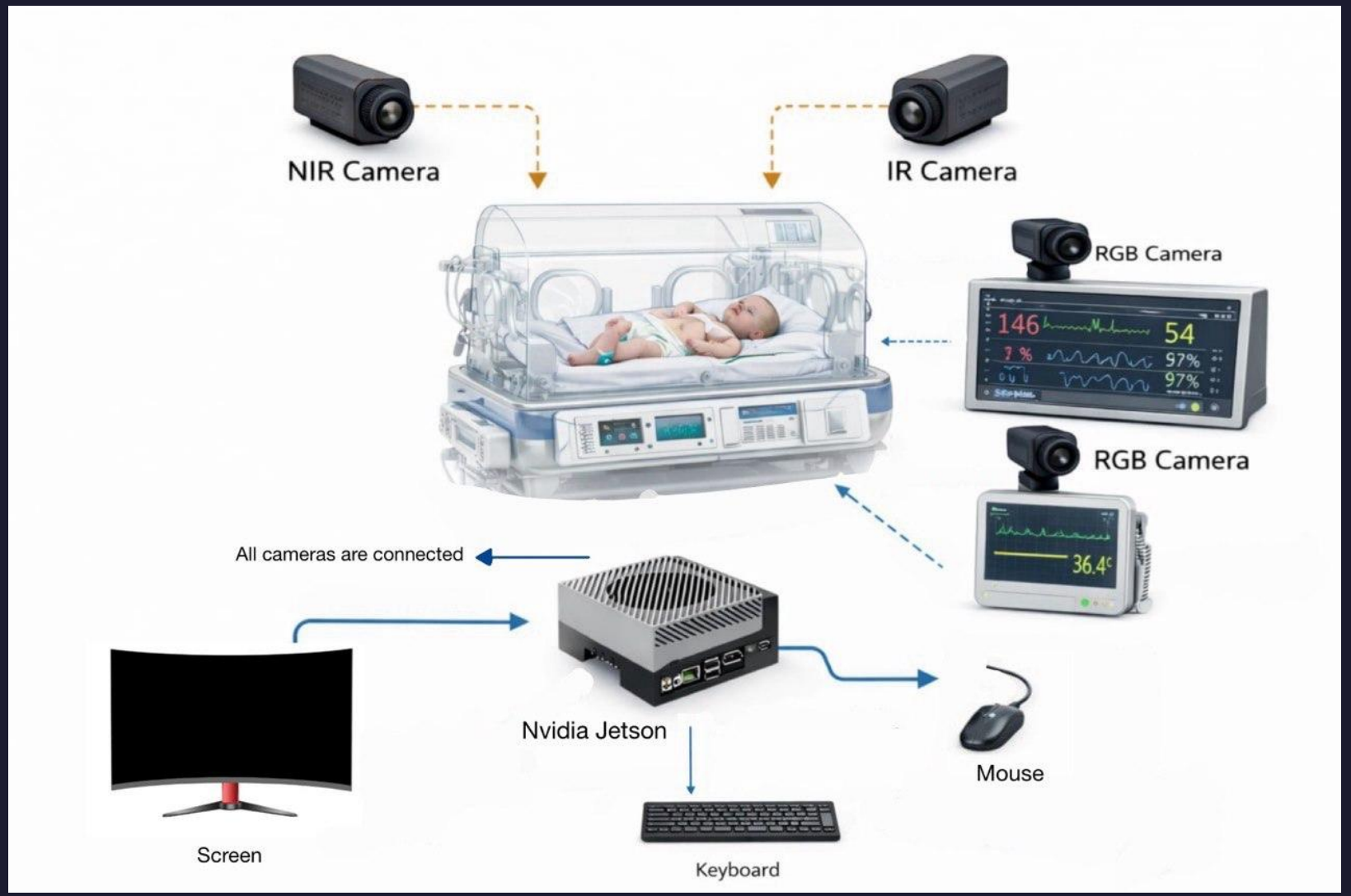
System Architecture



Experiments Performed

- Obtained Helsinki Committee (IRB) ethical approval for conducting non-contact video monitoring research on preterm infants.
- Conducted multiple clinical recording sessions at Soroka NICU, documented preterm infants under real clinical conditions.
- Purchased and integrated dual-camera system (RGB + NIR), NVIDIA Jetson AGX Orin edge compute, and storage hardware.
- 3D-printed custom mounts and articulating arms for incubator mounting.
- Created clinical data logging protocol capturing gestational age, weight, skin tone, and treatment metadata per session.
- Built GStreamer-based capture pipeline for synchronized, lossless recording at 30 fps.

System’s Setup



Interim Results

- Successful clinical data collection: multiple synchronized recording sessions completed at Soroka NICU.
- Monitor ROI Extraction: built a complete classical CV pipeline: camera calibration, lens undistortion, screen quad detection, KLT optical flow tracking, RANSAC homography estimation, Kalman-filtered corner smoothing, and perspective rectification.
- Ground Truth Pipeline: successfully isolating vital sign readings from medical monitor video feeds for training label generation.
- Object Detection Annotation: began annotating two classes (face, face+torso) using CVAT + Docker, structuring data in YOLO format for the Torso & Face Detection preprocessing stage.
- Project Status vs. Gantt Chart: Minor delay in early hardware phases, but aligned for final milestones. The preprocessing pipeline progress compensates for the initial setup overhead.

Challenges & Solutions

Frame Drops with Python/OpenCV

⚠ OpenCV VideoCapture at 30 fps dropped ~1 in 5 frames, breaking camera synchronization.

💡 Re-architected pipeline to GStreamer bash scripts, zero frame loss, stable sync.

Jetson Boot Failure

⚠ 100% memory utilization from accumulated video files caused complete boot failure.

💡 Connected via Ethernet/SSH, cleared space, installed & mounted 1TB NVMe SSD (/mnt/ssd), rerouted all pipelines.

Camera SDK & OS Compatibility

⚠ Hardware incompatibility: The purchased cameras do not support the Jetson's flashed OS, leading to broken SDK configurations across JetPack versions.

💡 Identified and flashed the latest compatible JetPack version supporting both camera SDK and Jetson AGX Orin hardware.

Conclusions & Future Work

Documentation Conclusions

- Adaptive Documentation: Refined our recording protocols dynamically to better suit the real-world constraints of the NICU environment.
- Deployment Learning: Adjusted camera positioning and settings to overcome lighting and space limitations in the clinical ward.
- Ground truth extraction pipeline validated on clinical monitor recordings.
- GStreamer-based recording ensures lossless, synchronized multi-camera capture at 30 fps.
- Clinical data logging protocol established for systematic dataset construction.

Remaining Tasks

HW & Documentation

- Ongoing NICU documentation at the hospital.

Architecture & Algorithms

- Integrate Video Motion Magnification (Autoencoder CNN) for RR branch.
- Train UNet for NIR → estimated IR conversion (Temperature branch).
- Build and train dual-branch Transformer architecture.
- Validate against ground truth. report MAE, RMSE metrics.